AI ASSISTANT CODING
ASSIGNMENT-18.2

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Batch: 34

Semester: VI

Branch: CSE(AIML)

Lab 18 – API Integration: Connecting to External Services with Error Handling

Task Description – 1 (Air Quality API Integration)

• Task:

Use AI assistance to develop a Python program that retrieves real-time air quality information for a specified city.

Instructions:

• Generate Python code using the requests library.

• Handle errors such as invalid city names, missing or incorrect API keys, and network timeouts.

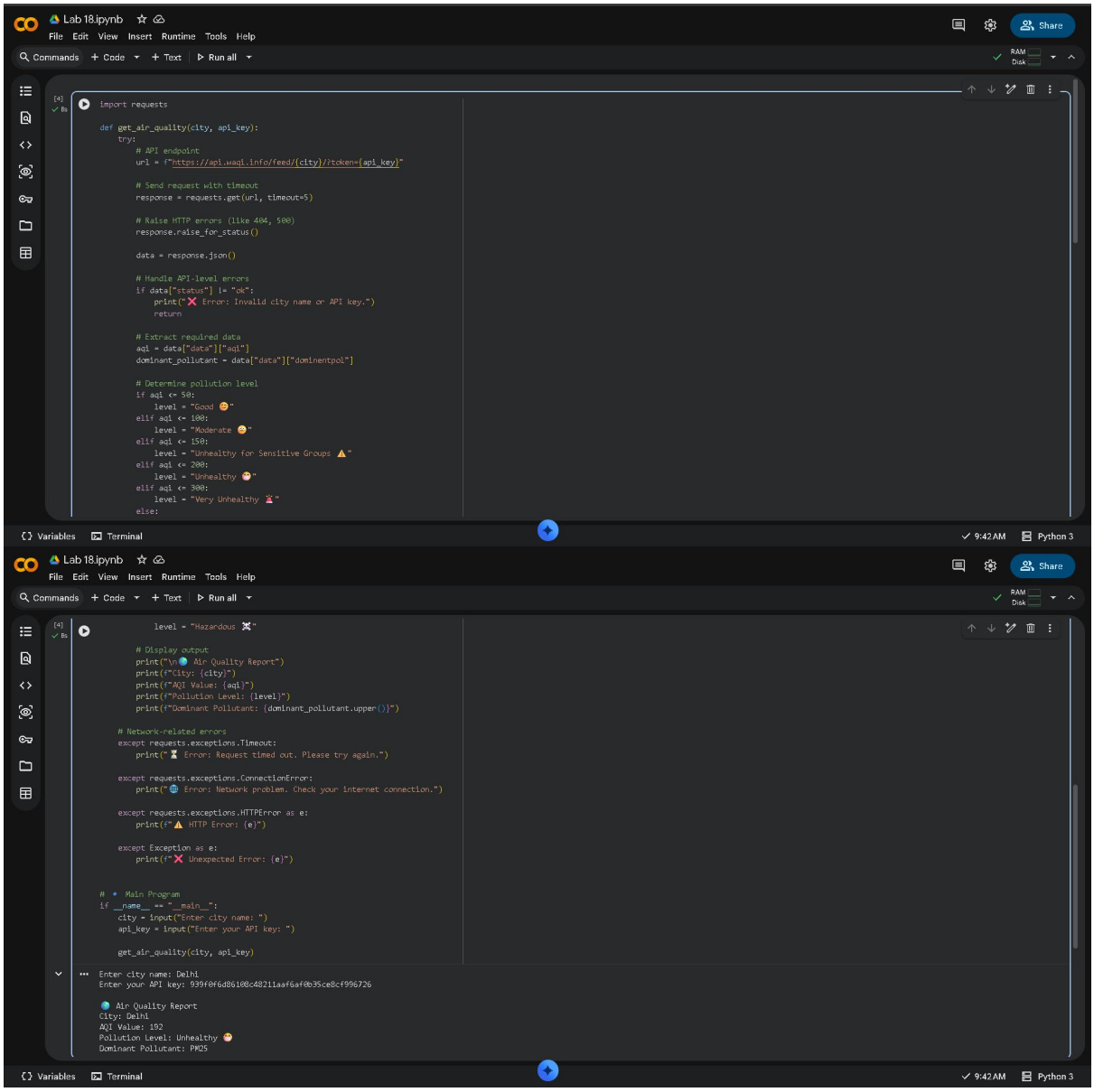
• Extract and display AQI value, pollution level, and dominant pollutant in a readable format.

Sample Code:

```
import requests

city = "Delhi"

url = "https://api.example.com/airquality"
```

Expected Output:

• A Python script that successfully fetches and displays air quality data with proper error handling.



Task Description – 2 (Language Translation API)

• Task:

Integrate a Language Translation API to translate text from English to two other languages.

Instructions:

• Use AI to generate API request code for translation.

• Validate user input text and target language codes.

• Handle API response errors and service unavailability.

• Display original and translated text clearly.

Sample Code:

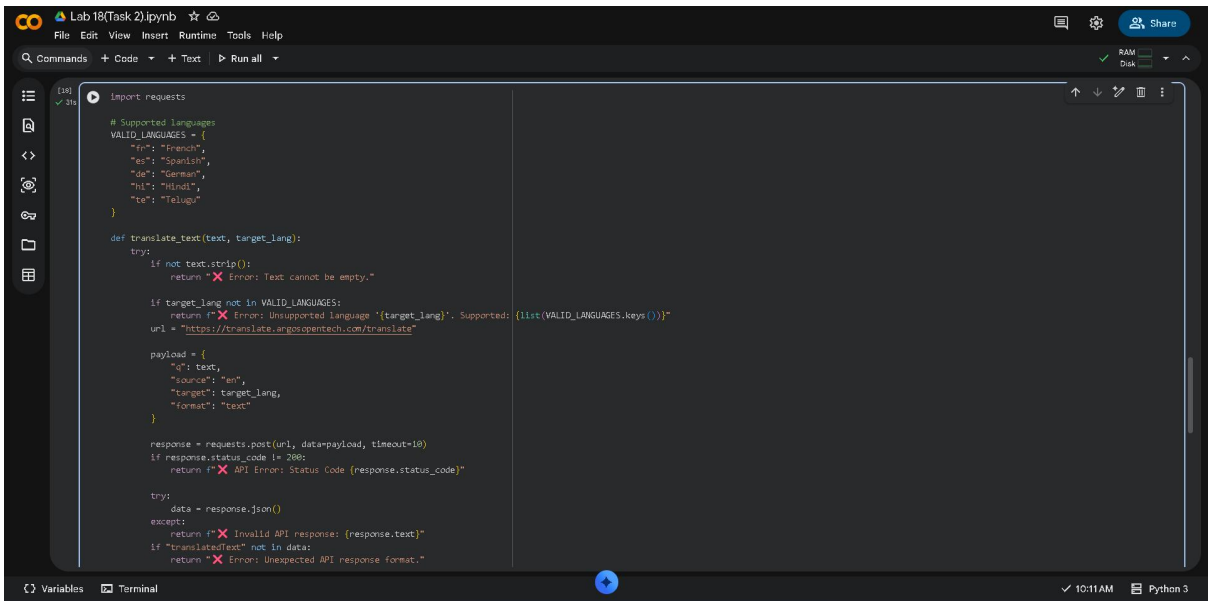
import requests

text = "Welcome to AI Assisted Coding"

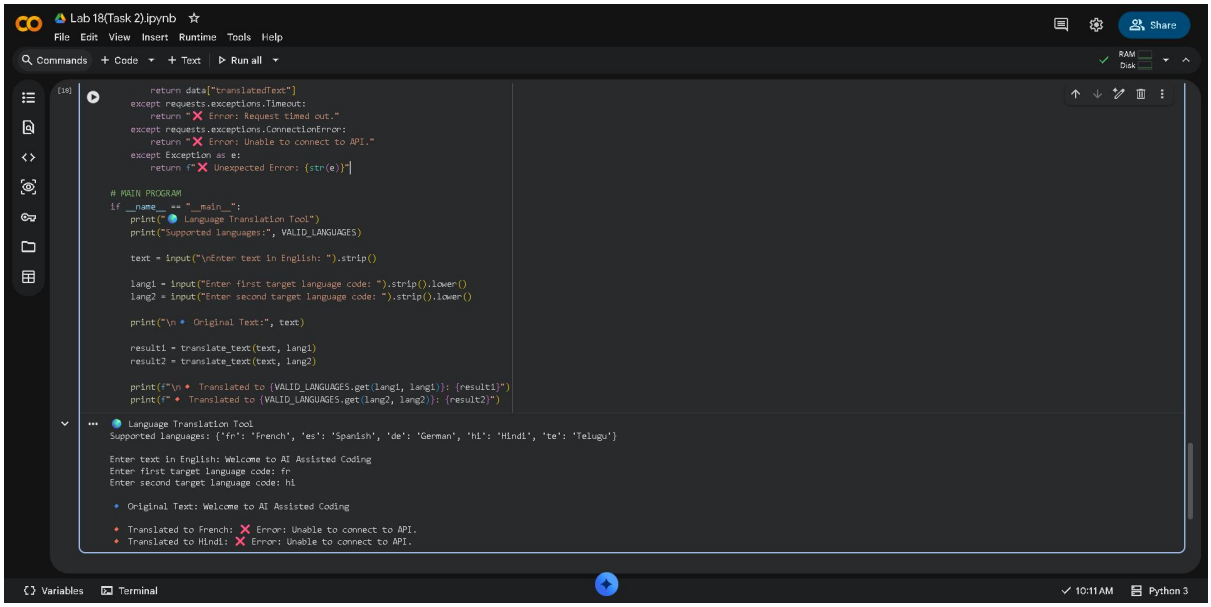
target_lang = "fr"

Expected Output:

• A working Python script that translates user input text with graceful error handling.



```
import requests

# Supported languages
VALID_LANGUAGES = {
    "fr": "French",
    "es": "Spanish",
    "de": "German",
    "hi": "Hindi",
    "te": "Telugu"
}

def translate_text(text, target_lang):
    try:
        if not text.strip():
            return "❌ Error: Text cannot be empty."

        if target_lang not in VALID_LANGUAGES:
            return f"❌ Error: Unsupported language '{target_lang}'. Supported: {list(VALID_LANGUAGES.keys())}"
        url = "https://translate.argosopentech.com/translate"

        payload = {
            "q": text,
            "source": "en",
            "target": target_lang,
            "format": "text"
        }

        response = requests.post(url, data=payload, timeout=10)
        if response.status_code != 200:
            return f"❌ API Error: Status Code {response.status_code}"

        try:
            data = response.json()
        except:
            return f"❌ Invalid API response: {response.text}"
        if "translatedText" not in data:
            return "❌ Error: Unexpected API response format."
```



```
        return data["translatedText"]
    except requests.exceptions.Timeout:
        return "❌ Error: Request timed out."
    except requests.exceptions.ConnectionError:
        return "❌ Error: Unable to connect to API."
    except Exception as e:
        return f"❌ Unexpected Error: {str(e)}"

# MAIN PROGRAM
if __name__ == "__main__":
    print("🌐 Language Translation Tool")
    print("Supported languages:", VALID_LANGUAGES)

    text = input("\nEnter text in English: ").strip()

    lang1 = input("Enter first target language code: ").strip().lower()
    lang2 = input("Enter second target language code: ").strip().lower()

    print("\n🔹 Original Text:", text)

    result1 = translate_text(text, lang1)
    result2 = translate_text(text, lang2)

    print(f"\n🔹 Translated to {VALID_LANGUAGES.get(lang1, lang1)}: {result1}")
    print(f"🔹 Translated to {VALID_LANGUAGES.get(lang2, lang2)}: {result2}")
```

```
🌐 Language Translation Tool
Supported languages: {'fr': 'French', 'es': 'Spanish', 'de': 'German', 'hi': 'Hindi', 'te': 'Telugu'}

Enter text in English: Welcome to AI Assisted Coding
Enter first target language code: fr
Enter second target language code: hi

🔹 Original Text: Welcome to AI Assisted Coding

🔹 Translated to French: ❌ Error: Unable to connect to API.
🔹 Translated to Hindi: ❌ Error: Unable to connect to API.
```

Task Description – 3 (Public Book Information API)

• Task:

Connect to a public Books API to fetch information about a book using its title or ISBN.

Instructions:

• Prompt AI to write Python code for sending GET requests to the API.

• Handle errors such as book not found, invalid input, and rate limits.

• Display book title, author, publisher, and publication year.

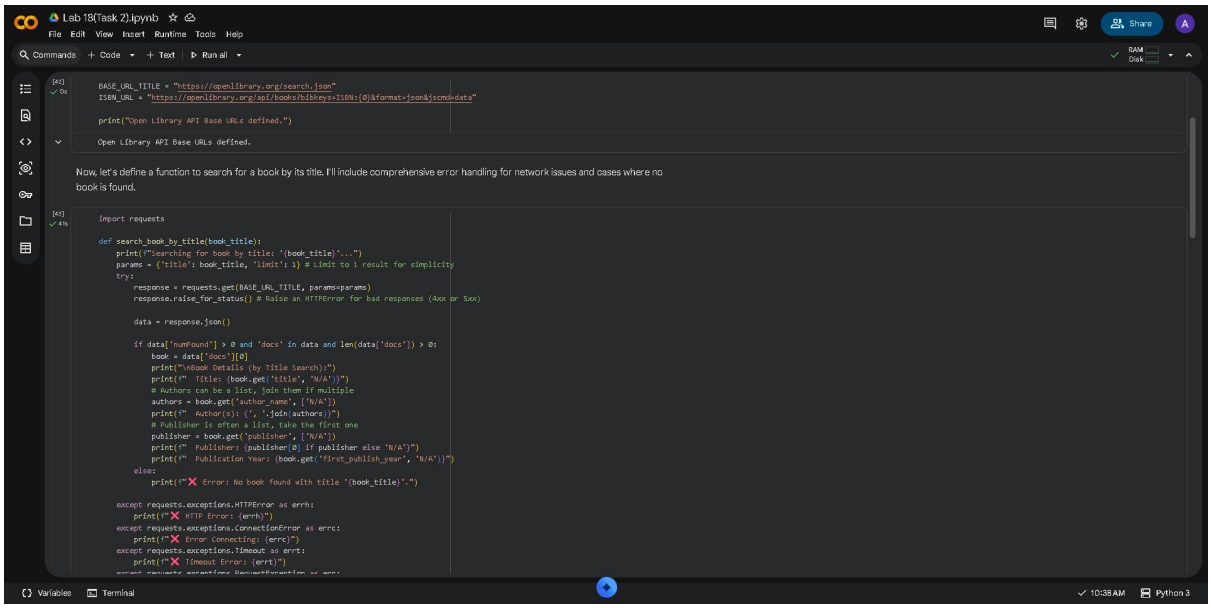
Sample Code:

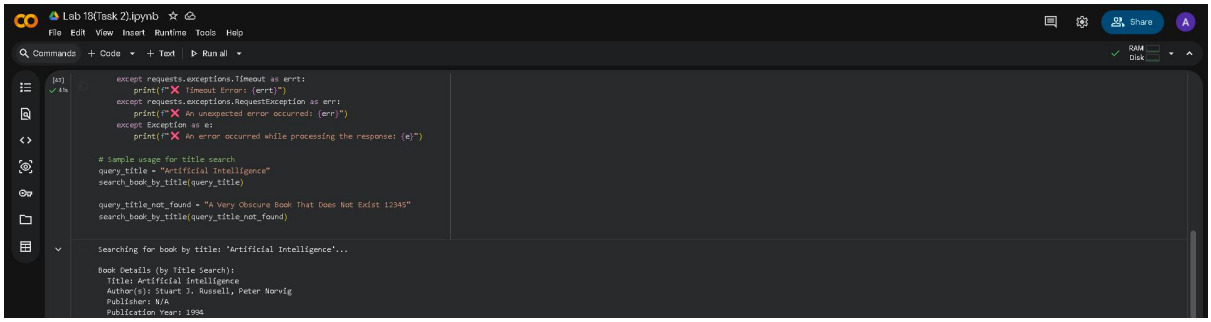
import requests

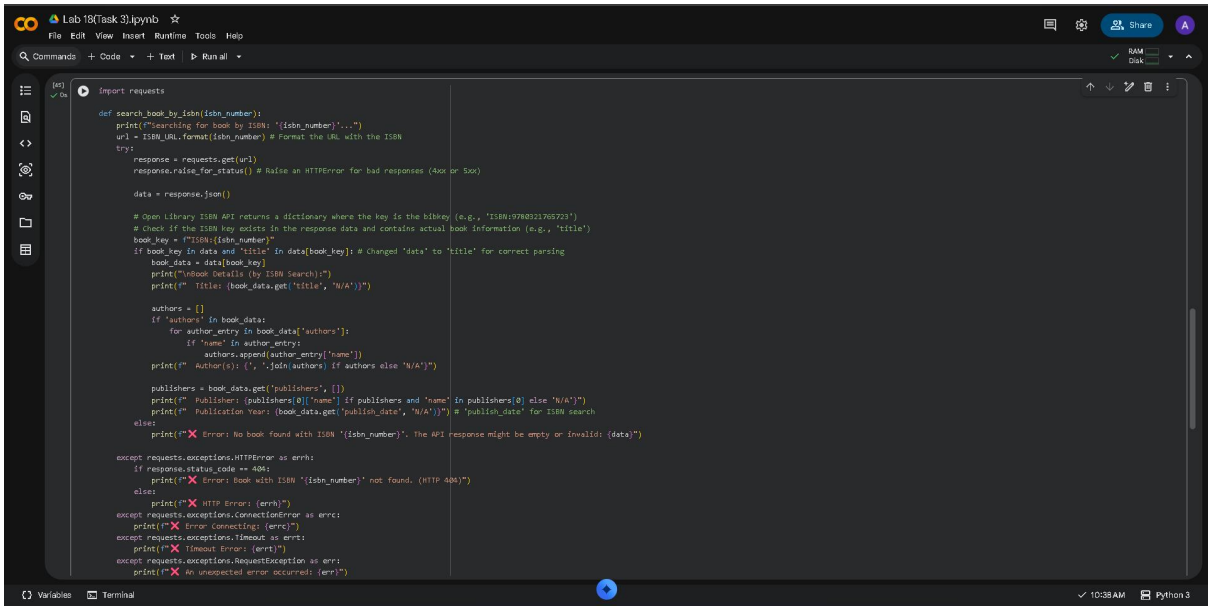
query = "Artificial Intelligence"

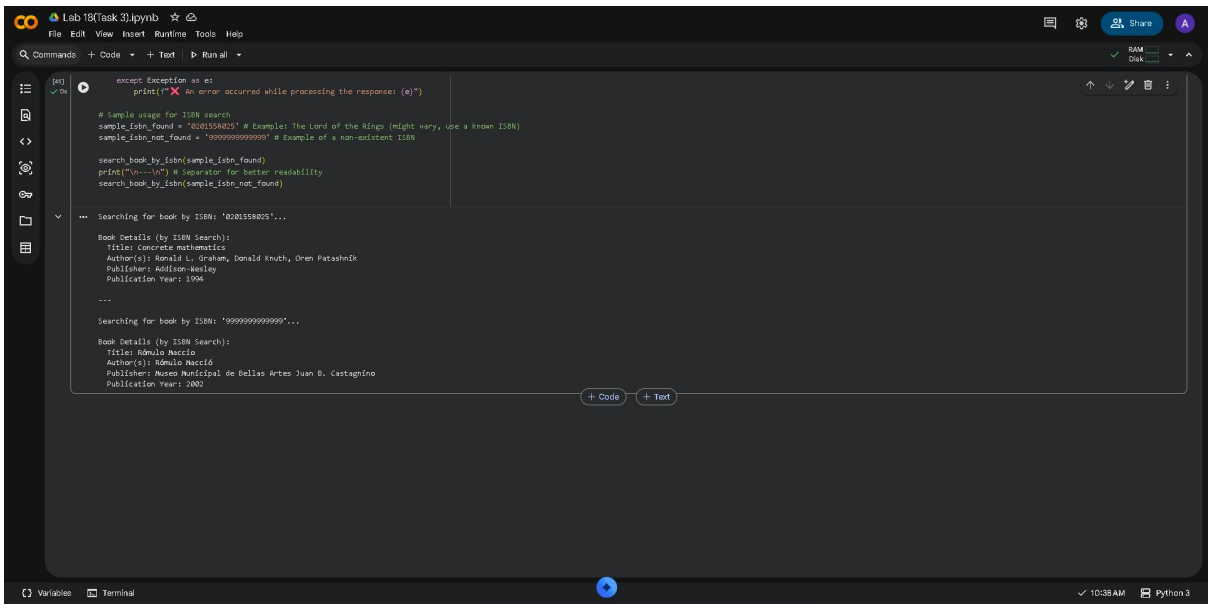
Expected Output:

• A Python program that retrieves and displays book details with robust error handling.









Task Description – 4 (Real-Time Application: Job Listings Aggregator)

• Scenario:

Develop a Job Listings Aggregator using a public Jobs API.

Requirements:

• Fetch the latest job postings for a given role or location.

• Handle errors such as invalid search terms, missing API keys, and request failures.

• Display job title, company name, location, and posting date in a numbered list.

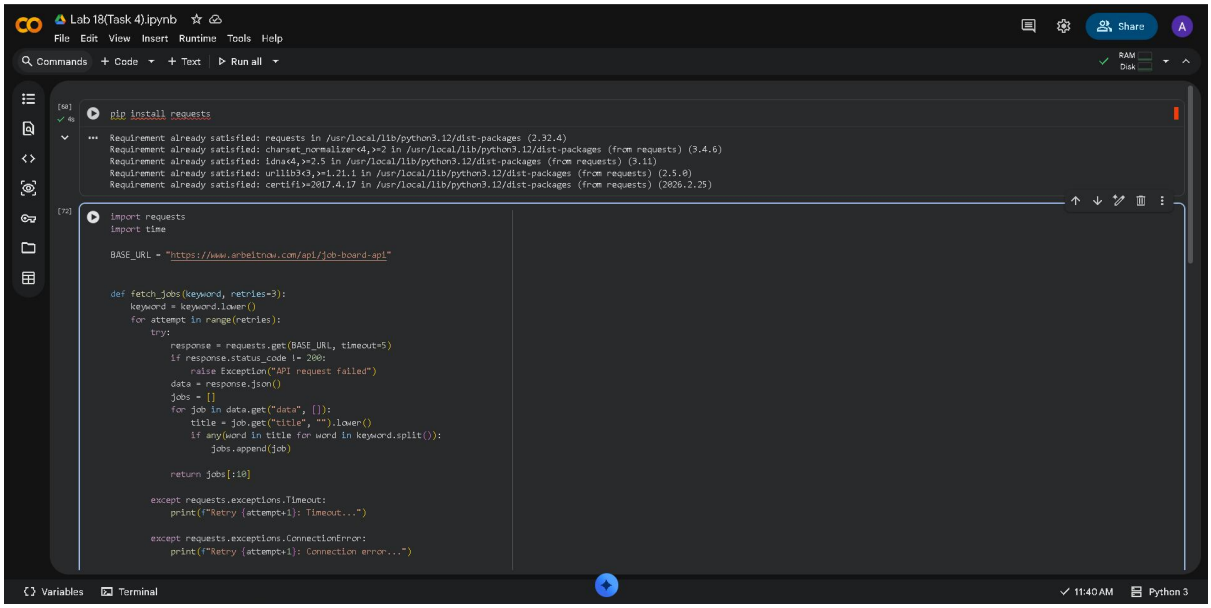
• Implement a retry mechanism for failed API requests.

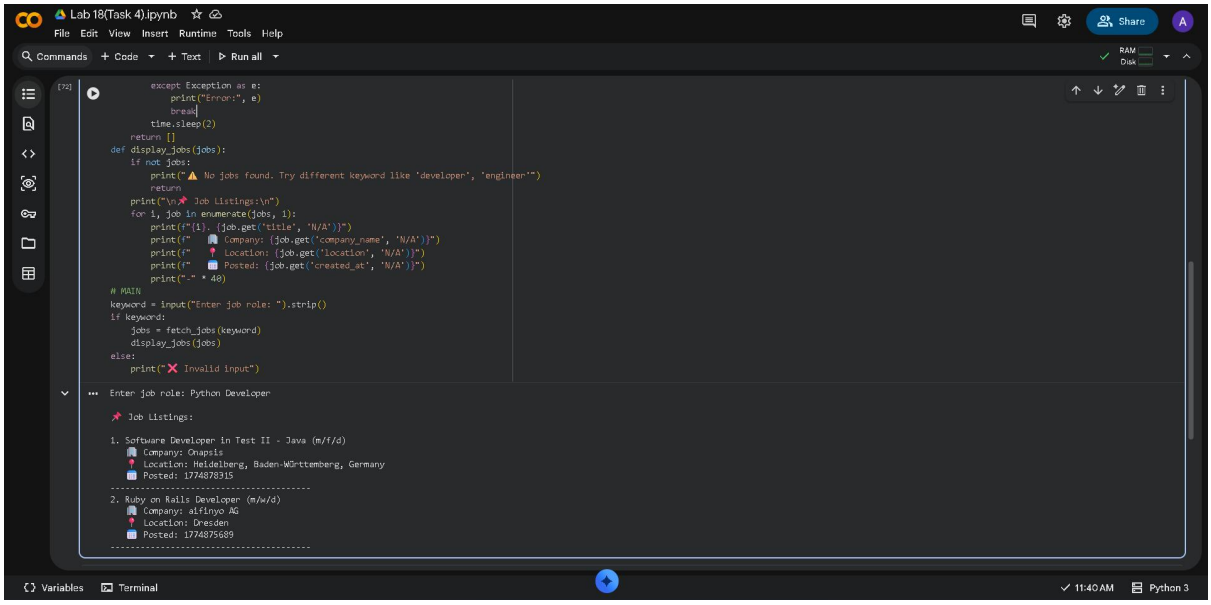
Sample Code:

```
import requests

keyword = "Python Developer"
```

Expected Output:

• A functional Python script that retrieves and displays job listings with proper error handling.





Task Description – 5 Stock Market Data API Integration

• Task:

Use AI assistance to develop a Python program that retrieves real-time stock market data for a selected company.

Instructions:

• Generate Python code using the requests library to connect to a public Stock Market API.

• Validate user input for stock symbol.

• Handle common errors such as invalid symbols, API rate limits, missing API keys, and network failures.

• Extract and display stock name, current price, day high, day low, and percentage change.

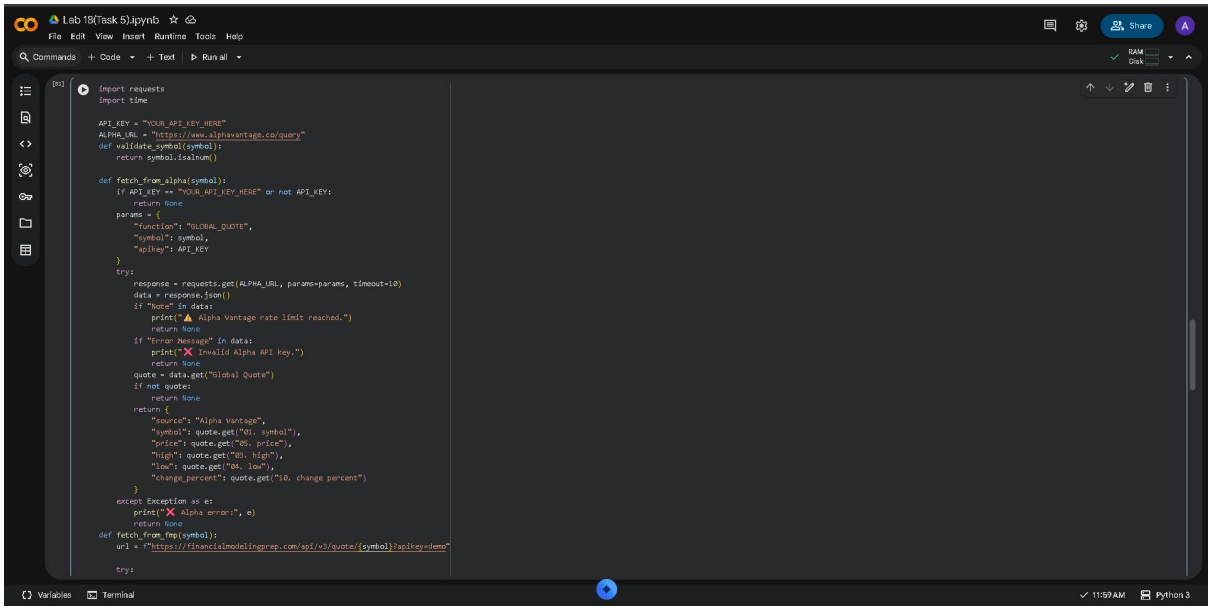
Sample Code:

import requests

symbol = "AAPL"

• Expected Output:

A Python script that successfully fetches and displays stock market information with robust error handling.



Lab 18(Task 5).ipynb

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[85]

```python
    try:
        response = requests.get(url, timeout=10)

        if response.status_code != 200:
            print("❌ API request failed.")
            return None
        data = response.json()
        if not data:
            print("❌ Invalid stock symbol.")
            return None

        stock = data[0]
        return {
            "source": "FMP API",
            "symbol": stock.get("symbol"),
            "price": stock.get("price"),
            "high": stock.get("dayHigh"),
            "low": stock.get("dayLow"),
            "change_percent": str(round(stock.get("changesPercentage", 0), 2)) + "%"
        }

    except requests.exceptions.Timeout:
        print("⏳ Request timed out.")
    except requests.exceptions.ConnectionError:
        print("🌐 Network error.")
    except Exception as e:
        print("❌ FMP error:", e)

    return None
def display_stock(data):
    print("\n📊 Stock Market Data")
    print("----------------------------")
    print(f"Source: {data['source']}")
    print(f"Symbol: {data['symbol']}")
    print(f"Current Price: {data['price']}")
    print(f"Day High: {data['high']}")
    print(f"Day Low: {data['low']}")
    print(f"Change: {data['change_percent']}")

def main():
    symbol = input("Enter stock symbol (e.g., AAPL, TSLA): ").strip().upper()
    if not validate_symbol(symbol):
        print("❌ Invalid symbol. Use only letters/numbers.")
```

Variables Terminal 11:59 AM Python 3

Lab 18(Task 5).ipynb

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[86]

```python
        return
    retries = 3
    for attempt in range(retries):
        print(f"\n🔄 Attempt {attempt + 1}...")
        # Try Alpha (optional)
        data = fetch_from_alpha(symbol)
        # Fallback to FMP (main working API)
        if not data:
            print("⚠️ Switching to FMP API...")
            data = fetch_from_fmp(symbol)
        if data:
            display_stock(data)
            return
        else:
            if attempt < retries - 1:
                print("🔄 Retrying in 2 seconds...")
                time.sleep(2)

    print("\n❌ All retry attempts failed.")
    print("⚠️ Possible issues:")
    print("   • No internet connection")
    print("   • API temporarily down")
    print("   • Invalid stock symbol")
if __name__ == "__main__":
    main()
```

```
Enter stock symbol (e.g., AAPL, TSLA): AAPL

🔄 Attempt 1...
⚠️ Switching to FMP API...
❌ API request failed.
🔄 Retrying in 2 seconds...

🔄 Attempt 2...
⚠️ Switching to FMP API...
❌ API request failed.
🔄 Retrying in 2 seconds...

🔄 Attempt 3...
⚠️ Switching to FMP API...
❌ API request failed.

❌ All retry attempts failed.
⚠️ Possible issues:
   • No internet connection
```

Variables Terminal 11:59 AM Python 3